2 The Earth may be considered to be a uniform sphere of radius 6.4×10^6 m and its mass is assumed to be concentrated at its centre.

(a) Show that the mass of the Earth is 6.0×10^{24} kg.

[1]

(b) A satellite of mass 650 kg is to be launched from the Equator and put into geostationary orbit.

(i) Determine the radius of the geostationary orbit.

radius = ……. m [3]

(ii) Calculate the increase in gravitational potential energy of the satellite during its launch from the Earth's surface to the geostationary orbit.

increase in gravitational potential energy = ……. J [2]

(c) Small resistive forces acting on the satellite cause the radius of its orbit to change. Explain why the kinetic energy of the satellite increases due to these resistive forces.

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…….....

……..

..... [3]

[1]

3 **(a)** State what is meant by *specific latent heat of vaporization*.